

Week 5: OpenAI, APIs, and Identity Integration

Week 5 established the essential foundation for building intelligent applications by focusing on connecting external AI intelligence via **OpenAI APIs** and securing access through **digital identity** management.

Core Curriculum

- **OpenAI API Mastery:** We learned the mechanics of integrating advanced language models. This involved understanding how to obtain and use **API keys**, specify model **endpoints**, and structure prompts to receive highly specific, structured JSON or text responses that could be immediately utilized within an application's logic.
- **API Authentication and Security:** A critical focus was placed on security. We studied the importance of managing **API keys** and **tokens** securely, not only for controlling access and preventing unauthorized use but also for adhering to rate limits and ensuring data protection.
- **Digital Identity and Personalization:** We explored how to handle **user identity**, implement secure **session management**, and verify authentication. This is crucial for maintaining user privacy and enabling the personalized, stateful experiences expected in modern AI applications.

Practical Application

We developed small, proof-of-concept programs that demonstrated dynamic AI interaction. These scripts used direct API calls to perform tasks such as content **summarization**, providing **code explanations**, and answering complex, **domain-related queries** based on user input. This solidified the process of integrating external intelligence with local application logic.